Synopsis

Behavioral Segmentation for Targeted Marketing

Problem Id: 10

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1. Problem Description:

The main aim of this project is to **analyze customer purchasing behavior and group similar customers together**. Businesses collect large amounts of customer data, but simply having the data does not always help them understand their customers. Therefore, this project focuses on dividing customers into different groups based on their **Annual Spending** and **Purchase Frequency**.

To achieve this, a **synthetic retail dataset** is created that represents customer purchasing patterns. The data is then preprocessed using **StandardScaler**, which helps bring all numerical values to a similar scale so that the analysis becomes fair and accurate. After preprocessing, the **K-Means clustering algorithm** is applied. This algorithm groups customers with similar spending habits into clusters.

To determine the optimal number of clusters, both the **Elbow Method** and **Silhouette Score** are used. Additionally, multiple feature combinations are tested during **Phase 3 feature selection** to identify the most effective variables for clustering.

The final model uses **Annual Income and Annual Spending**, as they provide the best balance between cluster separation and meaningful business interpretation.

In real life, businesses use this type of analysis to understand their customers better. It helps companies create **targeted marketing strategies**, offer personalized promotions, improve customer satisfaction, and increase sales by focusing on the right group of customers.

1.1 Dataset Description

The dataset used in this project is a **synthetically created retail dataset** that represents customer purchasing behavior in a realistic manner. It includes both numerical and categorical features that describe how customers spend and interact with retail systems.

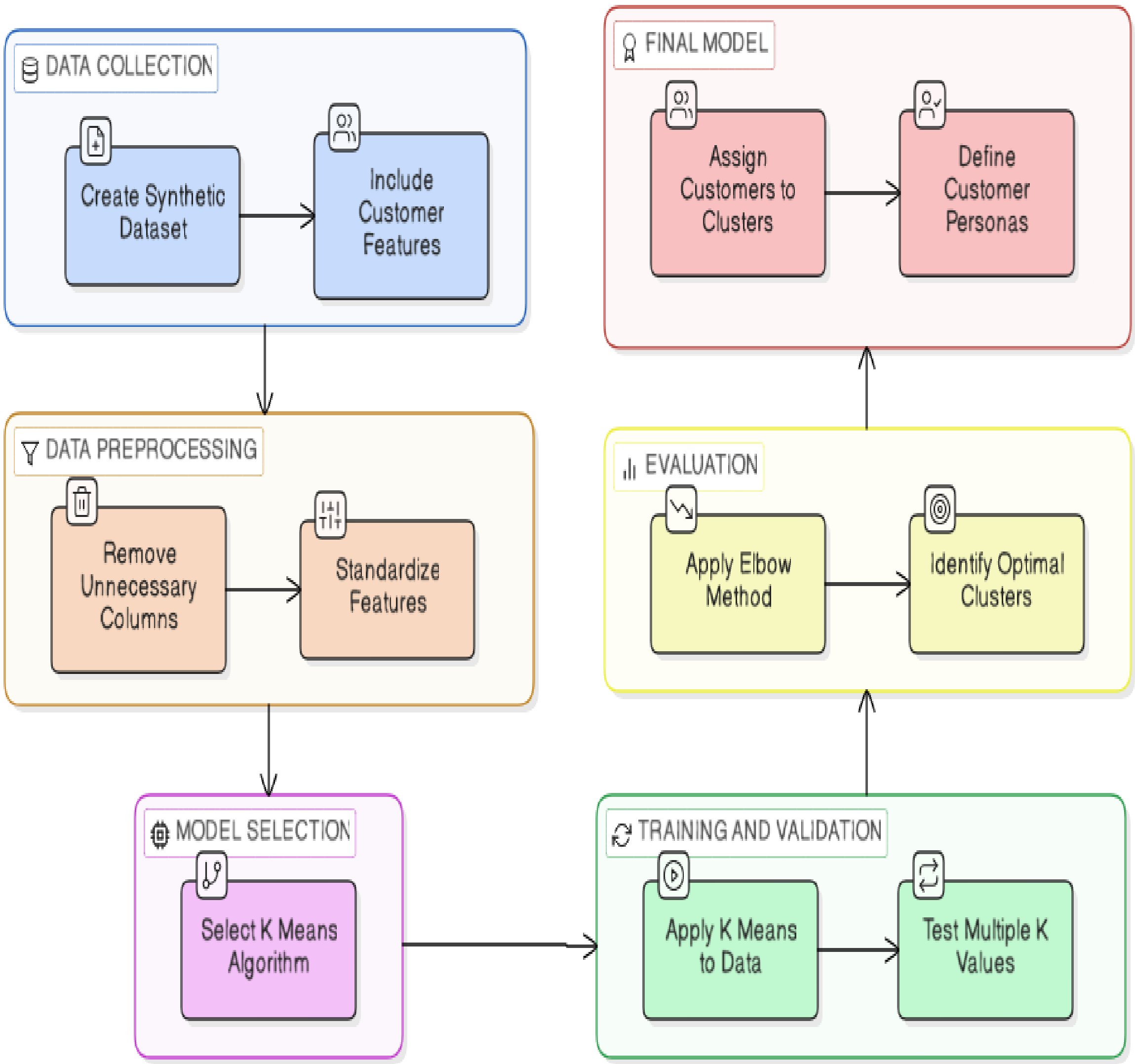
The main features are **Annual Income, Annual Spending, Purchase Frequency**, which help in understanding a customer's financial capacity and spending habits. These features are important for identifying patterns and grouping similar customers. Additional fields like **City and City Tier** are included to study how customer behavior varies across different locations and economic regions.

To make the dataset closer to real-world conditions, small variations and noise are introduced. This helps in practicing data preprocessing steps such as handling missing values, scaling features, and detecting outliers.

Overall, the dataset is suitable for building a complete machine learning pipeline, from data cleaning and preprocessing to applying K-Means clustering and generating useful insights for targeted marketing.

1.2 Steps of Implementation:

Fig. Architecture diagram



2. Modules / Libraries Used

1. pandas

Used for data manipulation and analysis. It helps in loading the dataset, handling missing values, selecting features, and organizing data in tabular format using DataFrames.

2. numpy

Provides support for numerical computations and array operations. It is useful for performing mathematical calculations and generating synthetic numerical data efficiently.

3. matplotlib

A visualization library used to create graphs and plots. In this project, it is mainly used to plot the **Elbow Method graph** and visualize clustering results.

4. seaborn

A statistical data visualization library built on top of matplotlib. It helps create more informative and visually appealing plots for exploratory data analysis.

5. scikit-learn (sklearn)

A machine learning library that provides tools for preprocessing and implementing machine learning algorithms.

6. sklearn.preprocessing.StandardScaler

Used to standardize numerical features so that they have a mean of 0 and a standard deviation of 1. This ensures that all features contribute equally during clustering.

7. sklearn.cluster.KMeans

Implements the **K-Means clustering algorithm** used to group customers into different segments based on similarities in their purchasing behavior

2.1 Evaluation metrics and Plots used:

1. Elbow Method (WCSS)

The Elbow Method will be used to determine the optimal number of clusters (K) for the K-Means algorithm by analyzing the reduction in within-cluster variance. It plots the **within-cluster sum of squared** distances (inertia) against different values of K. The point where the graph forms an "elbow" indicates the most suitable number of clusters.

2. Scatter Plot for Cluster Visualization

A scatter plot will be used to visualize the clusters formed by the K-Means algorithm. The x-axis will represent Annual Spending, and the y-axis will represent Purchase Frequency. Different colors will represent different customer clusters, helping to clearly observe customer segments.

3. Silhouette Score

Measures how well-separated the clusters are.

- Value close to 1 → well-defined clusters
- Value near 0 → overlapping clusters

4. Cluster Distribution Plot

A simple plot will be used to show the number of customers in each cluster. This helps in understanding how customers are distributed among different groups.

2.2 Platform Used:

Jupyter Notebook & Google Collab is used to run the machine learning pipeline in an organized manner. Each stage of the process such as **loading the dataset, preprocessing the data, applying StandardScaler, running the K-Means algorithm, and visualizing results using plots** can be executed in separate cells.

3. Machine Learning Models Used:

o K-Means Clustering

K-Means Clustering is an iterative, unsupervised machine learning algorithm that partitions a dataset into k non-overlapping clusters based on feature similarity. It operates by minimizing the Within Cluster Sum of Squares (WCSS), ensuring that data points within each cluster are as close as possible to their respective centroids (the mean of all points in the cluster) while maximizing the distance between different clusters.

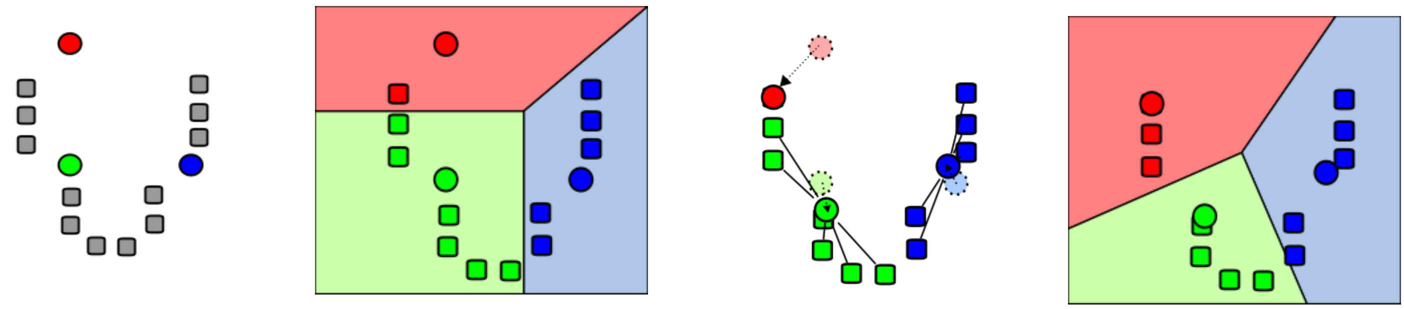
Key Characteristics:

- Uses Euclidean distance
- Requires predefined number of clusters
- Iteratively updates centroids

In This Project:

- Multiple feature sets were tested
- Best features selected using silhouette score
- Final model uses **Income + Spending (k = 4)**

Demonstration of the standard algorithm



1. k initial "means" (in this case k=3) are randomly generated within the data domain (shown in color)

2. k clusters are created by associating every observation with the nearest mean. The partitions here represent the Voronoi diagram generated by the means.

3. The centroid of each of the k clusters becomes the new mean.

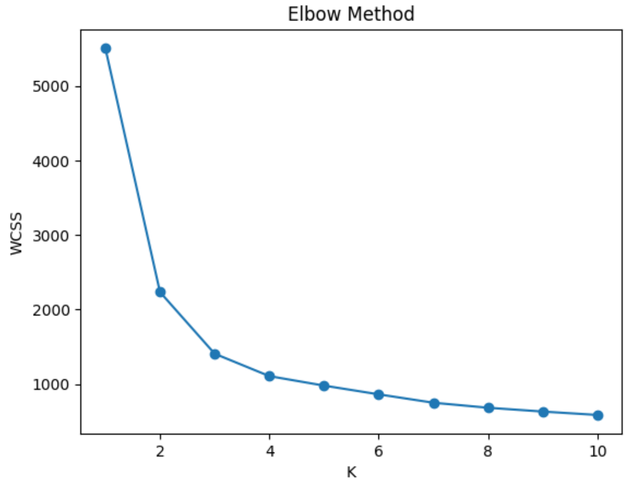
4. Steps 2 and 3 are repeated until convergence has been reached.

4. Evaluations and Visualizations

The results of the project demonstrate the effectiveness of K-Means clustering in segmenting customers based on their financial capacity and spending behavior.

a. Elbow Method Analysis

The Elbow Method was used to determine the optimal number of clusters. The graph showed a clear bend around **k = 3**, indicating that increasing clusters beyond this point does not significantly improve model performance.



b. Silhouette Score Evaluation

Multiple values of k were tested using the Silhouette Score. Although the highest score was observed at **k = 2**, it resulted in overly generalized clusters. Therefore, **k = 3** was selected as it provides a better balance between cluster quality and meaningful segmentation.

```
Silhouette Score :
K:2 -> 0.472
K:3 -> 0.406
K:4 -> 0.353
K:5 -> 0.299
K:6 -> 0.303
K:7 -> 0.299
K:8 -> 0.286
K:9 -> 0.283
K:10 -> 0.277
```

c. Feature Selection Results

Different feature combinations were tested:

- Income + Frequency
- Spending + Frequency + Score
- Income + Spending + Frequency
- Income + Spending (Final Selection)

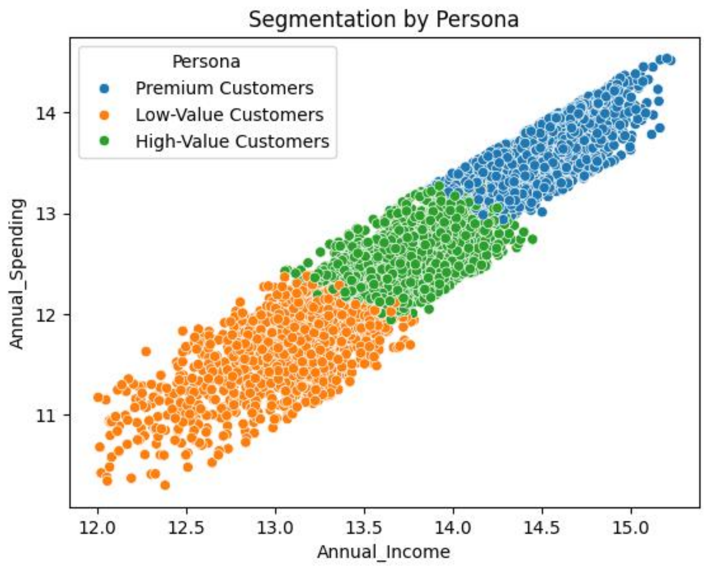
The combination of **Annual Income and Annual Spending** produced the best clustering performance among valid multi-feature sets. Our selected pair performed the best.

```
Feature Pair Comparison (k=3):
Rank | Silhouette | Features
--------------------------------------------------
   1 |     0.4915 | Annual_Income + Annual_Spending <-- our choice
   2 |     0.4243 | Annual_Spending + Spending_Score
   3 |     0.3723 | Purchase_Frequency + Spending_Score
   4 |     0.3595 | Annual_Income + Spending_Score
   5 |     0.3560 | Annual_Income + Purchase_Frequency
   6 |     0.3418 | Annual_Spending + Purchase_Frequency
```

```
Best pair : ('Annual_Income', 'Annual_Spending')
```

d. Cluster Visualization

Scatter plots were used to visualize clusters using **Annual Income (x-axis)** and **Annual Spending (y-axis)**. The clusters showed clear separation, indicating effective grouping of customers into distinct segments.



e. Cluster Profiling

Each cluster was analyzed using mean values of Income and Spending:

- High (Income + Spending) → **Premium Customers**
- Moderate (Income + Spending) → **High-Value Customers**
- Low (Income + Spending) → **Low-Value Customers**

f. Cluster Distribution

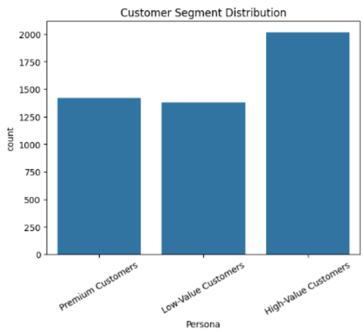
The distribution of customers across segments is as follows:

Persona	count
High-Value Customers	2014
Premium Customers	1422
Low-Value Customers	1377

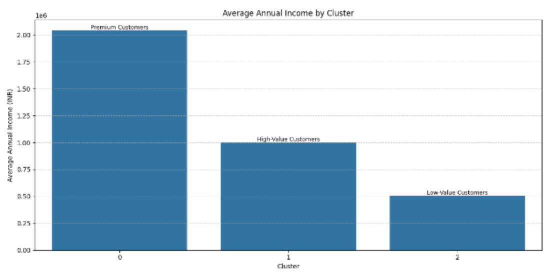
This shows that most customers belong to mid-to-high value segments, indicating a stable and profitable customer base.

g. Other Graphs

1.City_Tier Distribution



2.Persona Distribution



3.Average Income by Cluster

6. Conclusion and Future Scope

Conclusion

This project successfully demonstrates customer segmentation using the K-Means clustering algorithm on a retail dataset. The data was preprocessed using StandardScaler to ensure fair contribution of all features, and multiple feature combinations were evaluated during Phase 3.

The feature selection process showed that **Annual Income** and **Annual Spending** provide the most effective clustering results. Although simpler or derived features such as Spending Score produced higher silhouette values, they were not selected because they lack multidimensional interpretability.

The optimal number of clusters was chosen as **k = 3**, as it provides a good balance between cluster quality and meaningful segmentation. The resulting clusters represent distinct customer personas such as high-value, premium, regular, and low-value customers.

Overall, the model successfully converts raw customer data into actionable insights that can support targeted marketing strategies and improve business decision-making.

Future Scope

- Use **real-world datasets** to improve reliability and applicability
- Apply advanced clustering techniques such as **DBSCAN or Hierarchical Clustering**
- Incorporate additional features like customer demographics and behavioral history
- Develop an **interactive dashboard** for real-time segmentation and analysis
- Integrate with recommendation systems for **personalized marketing strategies**

7. Books and Link Sources:

- Dataset Link: Click

1. Artificial Intelligence: A Modern Approach ~ Stuart Russell & Peter Norvig
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow ~ Aurélien Géron
3. Python for Data Analysis ~ Wes McKinney
4. Introduction to Machine Learning with Python ~ Andreas C. Müller & Sarah Guido

- https://scikit-learn.org
- https://pandas.pydata.org
- https://matplotlib.org
- https://scholar.google.com